

Nate Cirino

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Available May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Digital Design & Computer Architecture, Computer Systems, Algorithms & Data Structures, Object-Oriented Design, Computer Networks, Foundations of Software Engineering, Foundations of Data Science, Systems Security, Cybersecurity

PROJECTS

Kepler | *Rust, C++, Tokio (async), Raft, LSM-tree, MVCC*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Hand-built the low-level storage path – in-memory memtable, write-ahead log, and leveled on-disk SSTables – plus Raft election, log replication, and membership changes
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across multi-node Raft cluster
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution on the hot write path, with asynchronous I/O and task scheduling on Tokio

NOAA AUV Sonar Detection | *Python, C, PyTorch, AWS Neuron SDK, SageMaker*

April 2025 – July 2025

- Trained CNN models over thermal-sensor spectrogram imagery to detect novel aquatic signatures for NOAA AUVs
- Refined architecture and loss formulation to address class imbalance and sensor noise in the image pipeline, improving detection precision on degraded inputs
- Designed AWS Neuron SDK deployment path for onboard custom-silicon accelerator inference under hard real-time latency limits, trading precision against per-frame compute budget

Sentinel | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker*

Apr. 2026 – Present

- Built an AI SRE agent autonomously triaging incidents via tool-use over Prometheus, Tempo, and Loki telemetry
- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Designing a supplier-agnostic e-label API against an open-ended spec, unifying 4 vendor backends behind one interface and powering new outlet launches
- Own front-of-house device stack across all Wayfair physical retail brands (AllModern, Birch Lane, Joss & Main, Perigold): e-labels, mobile, and customer/associate-facing POS
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Root-caused and overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating latency timeouts at production scale
- Profiled and optimized Expert AI search and filtering across 20+ microservices, improving clinical query relevance at scale
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow
- Integrated Expert AI into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Made architectural calls on an unspecified greenfield stack, then led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows
- Architected Flask backend with RAG pipeline for personalized workout recommendations using Bedrock KB
- Evolved RAG stack from LangChain to Vespa to zero-ops Bedrock KB, improving retrieval and reducing overhead
- Maintained 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Continue as technical advisor on architecture and agentic-workflow direction

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

SKILLS & HONORS

Languages/Tools: C++, C, Rust, Python, Java, TypeScript, JavaScript, Bash, Linux, Docker, Kubernetes, AWS, Azure, Terraform, Jenkins, Buildkite, Groovy, PostgreSQL, GraphQL

Frameworks & Libraries: PyTorch, TensorFlow, Tokio, Pandas, FastAPI, Flask, Node.js, Express, React, Next.js

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor